RAG SYSTEM FOR EXCHANGE PROGRAM Q&A

Meichen Dong

TABLE OF CONTENT

- Overview & Motivation 01
- System Architecture 02
- Experiments & Evaluation 03
- Key Results at a Glance 04
- Prompt Template Experiments 05
- Gradio Interface 06
- Conclusion 07

OVERVIEW & MOTIVATION

There are 8 official exchange program PDFs, Students must read everything manually.
Information scattered across files, Hard to compare universities
Time-consuming and error-prone

➡ **build a Retrieval-Augmented Generation (RAG) system**

Users can ask questions in natural language and receive concise answers based on the official program materials.

SYSTEM ARCHITECTURE

01 PDF Loading

02 Chunking

03 Embedding

04 ChromaDB

05 LLM

06 Gradio UI

Query Flow

User question → Embed (MiniLM) → Retrieve top-k chunks → Build prompt → LLM answer → Display in Gradio

EXPERIMENTS & EVALUATION

6.1 RAG vs. No RAG	Without retrieval the LLM hallucinates. RAG is essential.
6.2 Embedding Models	all-MiniLM-L6-v2 outperforms paraphrase-MiniLM-L3-v2
6.3 Top-k Chunks	k = 4 gives the best balance; beyond 4 adds noise
6.4 Response Time	Average end-to-end latency ~2.3 s per query
6.5 LLM Comparison	Qwen2.5-72B gives more complete answers than zephyr-7b
6.6 Prompt Templates	Role-based > Structured > Basic
6.7 Accuracy	75% accuracy — 6 of 8 questions correct.

KEY RESULTS AT A GLANCE

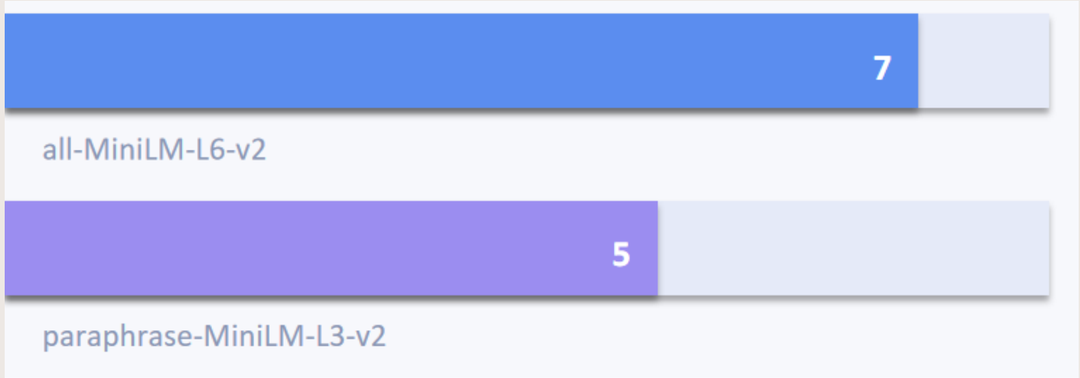
Accuracy
75%

Avg. response
2.3s

Best chunk count
k = 4

Correct answers
6 / 8

Embedding comparison — retrieved hits (8 test queries)



Answer quality by top-k (1–5 scale, estimated)

PROMPT TEMPLATE EXPERIMENTS (6.6)

BASIC

Simple but unreliable — LLM sometimes ignores context.

```
Context: {context}

Question: {question}

Answer:
```

Baseline

STRUCTURED (XML)

XML tags help the model distinguish context from query.

```
<context>
{context}
</context>

<question>
{question}
</question>

Answer:
```

Better

ROLE-BASED

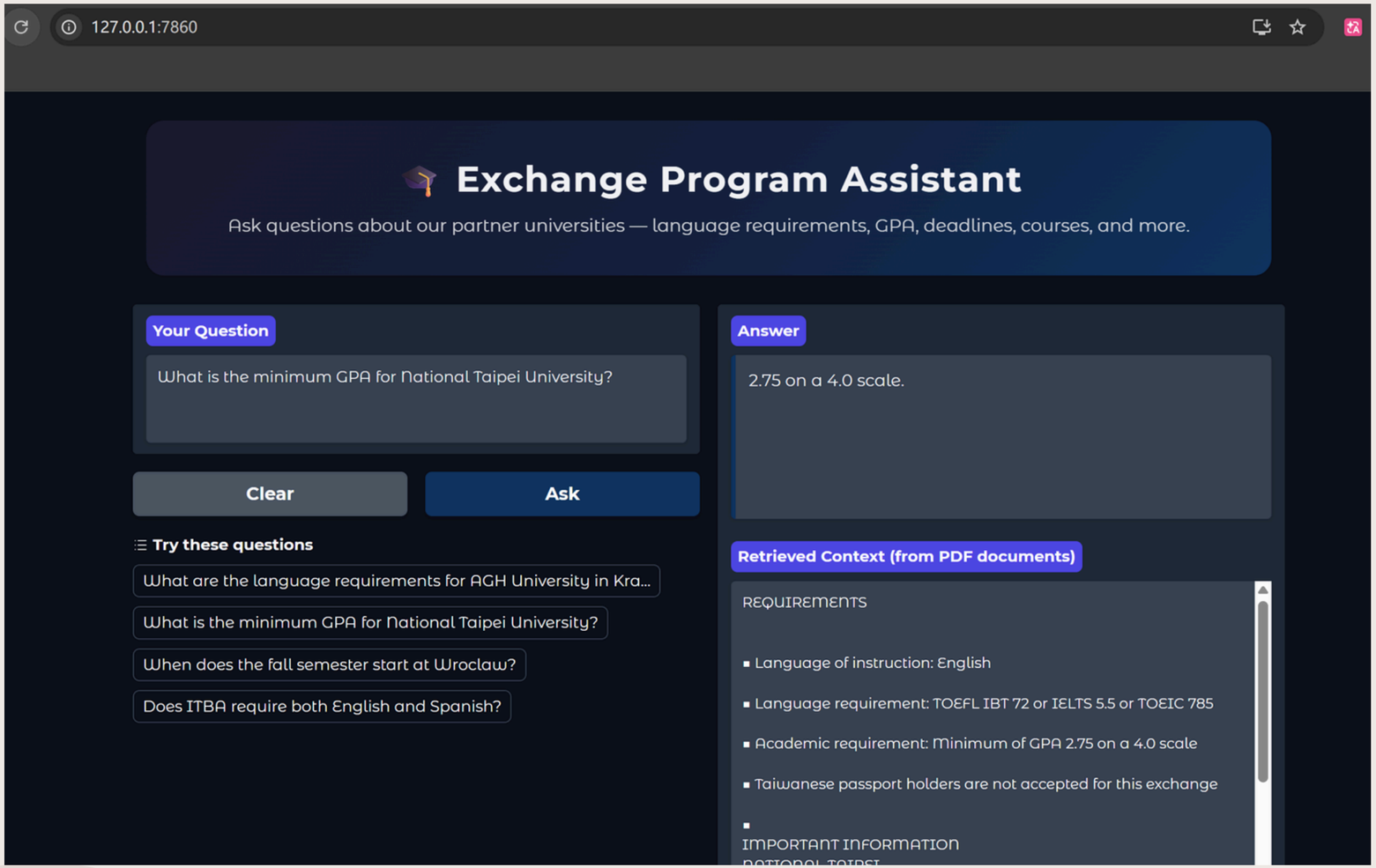
Best results — role framing produces precise answers.

```
You are an advisor for BSc
exchange students. Use only
the context below.

Context: {context}
Q: {question}
Answer:
```

Best !!

GRADIO INTERFACE



CONCLUSION

- **RAG pipeline successfully retrieves relevant information from 8 exchange PDFs**
- **ChromaDB PersistentClient enables fast, persistent semantic search on disk**
- **Role-based prompt template consistently produces the most accurate answers**
- **75% accuracy — 6 of 8 manually labelled questions answered correctly**
- **Gradio interface makes the system accessible without technical knowledge**

Future work: fine-tuned embeddings, larger evaluation set, multilingual support

THANK YOU